

Study of NGC 6946 galaxy star formation rate with optical and spectroscopic measurements.

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ABSTRACT

The aim of this report is to compute the star formation rate of the galaxy NGC 6946 and compare it to the values available in the literature. The star formation rate is derived from observations of flux in the H-alpha filter, corrected for dust attenuation using a spectroscopic analysis. The position of the galaxy in the BPT diagram is also taken into account. The value obtained for the star formation rate is higher than expected.

Key words: star formation rate – NGC 6946 – optical observations – spectroscopy

1 INTRODUCTION

The star formation rate (*SFR*) quantifies how many stars a galaxy forms per year. Galaxies can be classified as active or passive based on their *SFR*: a high value indicates a star-forming system, whereas a passive galaxy exhibits a lower rate. The *SFR* value varies among galaxies depending on several factors, including the environment (e.g. interactions with other galaxies, external gas accretion), the presence of feedback processes and also the cosmic epoch in which galaxies form. There are different methods to estimate the *SFR*. One of the most used relies on the H-alpha ($H\alpha$) emission line: ionizing photons produced by massive ($\gtrsim 10M_{\odot}$) short-lived stars can excite electronic transitions in hydrogen atoms, which subsequently undergo recombination processes, leading to the emission of recombination lines. In particular, the $H\alpha$ line corresponds to the $3 \rightarrow 2$ recombination of the hydrogen atom at $656.28nm$. This is consequently a reliable observable tracer of recent star formation activity. However, dust within the galaxy absorbs part of this emitted flux. So, it is essential to take this process into account. In particular, dust re-emits the absorbed energy at longer wavelengths and the Balmer decrement method, based on the observed $H\alpha/H\beta$ ratio, provides a way to quantify the attenuation and recover the intrinsic $H\alpha$ flux.

This report focuses on the galaxy NGC 6946. Being a face-on, massive and active galaxy, it is ideal for studying its star forming regions and their properties. This galaxy is also known as "Firework Galaxy" and is classified as starburst due to its high number of supernova events. According to the literature, it lies at a distance $D = (7.72 \pm 0.32)Mpc$ with a theoretical star formation rate of approximately $SFR \approx 3 \frac{M_{\odot}}{yr}$, based on respectively Kessler et al. (2020) and PHANGS- $H\alpha$ catalog (Leroy et al. (2019)). In addition, the analysis by Kessler et al. (2020) plays a central role in quantifying both the dust content and its spatial distribution within this galaxy.

The Baldwin-Phillips-Terlevich (BPT) diagnostic diagram is particularly effective for analyzing possible contamination in the ionization mechanisms coming from non-stellar sources, like active galactic nuclei. The key quantities are emission line flux ratios such as $\frac{NII}{H\alpha}$ and $\frac{OIII}{H\beta}$. By comparing them with the theoretical and empirical curves by Kewley et al. (2001), Kauffmann et al. (2003) and Draine (2003), the dominant ionization mechanism can be identified.

2 OBSERVATIONS AND DATA REDUCTION

2.1 Observations and datasets

The data used in this work are optical images obtained with the Bicocca Telescope (TOBi), located on the roof of the U9 building at University of Milano-Bicocca. The telescope adopts a Riccardi-Dall Kirkham optical design with a primary mirror and a focal ratio of 7, with two field flatteners. The flux is collected using a Zeus 445M Pro camera with full-frame SONY CMOS sensors, giving a resolution of 9576×6388 pixels with a scale of $0.276''$. The telescope is equipped with a filter wheel, and in this observation the H-alpha and r-band are used. They peak at $656nm$ and $658nm$ respectively with a bandwidth of $3.5nm$ and $\sim 140nm$. The optical data were collected over two nights: 02/10/2025 for the $H\alpha$ filter and 29/09/2025 for the r-band. In both cases, the galaxy was at an elevation of about 75° , roughly 30° over the practical horizon of Milan (considered at $\sim 45^\circ$). For the H-alpha filter, three exposures of 1200 seconds were acquired, for a total integration time of 60 minutes. While for the r-band filter, four exposures of 300 seconds each were obtained. The exposure times for both filters are sufficient to achieve a signal-to-noise ratio (S/N) greater than 10 in a chosen sample region of radius $2''$, ensuring that the galaxy image is sufficiently resolved above noise contributions. The S/N value is computed using Equation 1, assuming a constant surface brightness across the galaxy, obtained from the theoretical value of *SFR* (Leroy et al. (2019)).

$$\frac{S}{N} = \frac{n_* t}{\sqrt{n_* t + N_{Pix} \times (n_{Sky} t + n_{Dark} t + N_{RON}^2)}}. \quad (1)$$

In the expression, n_* represents the total number of electrons produced by the source, while the denominator includes all noise sources: the source itself (n_*), the sky contribution (n_{Sky}), the dark current (n_{Dark}) and the readout noise (N_{RON}). All these quantities are defined per pixel, so they are multiplied by the total number of pixels N_{Pix} to obtain the total contribution.

Spectroscopic data are also used, taken with the Hershel telescope using the ISIS spectrograph. Lastly, for the BPT diagram, a sample of 10000 galaxies from the Sloan Digital Sky Survey (SDSS), an optical and infrared spectroscopy database of the sky, is employed.

2.2 Data reduction

2.2.1 Telescope noise

All images are subjected to several noise sources that have to be taken into account before analyzing any physical quantities. In particular, the quantification of flux intensities is achieved through the measurement of photoelectrons extracted from the CMOS detector via photoelectric effect. However, intrinsic noise mechanisms within the camera can contribute additional electrons, potentially leading to an overestimation of the source flux.

The primary noise contributions include readout noise (RON), dark current, flat-field, and bias noise. Readout noise originates from statistical fluctuations during electron transport process within the readout electronics. Even if CMOS technology typically exhibits lower RON, this noise source is still present. It cannot be removed directly, but since bias frame fluctuations arise from RON noise, subtracting the bias also removes this contribution. The bias is a baseline signal always present in the telescope introduced by electronic offset voltages during the signal processing, necessary to maintain the detector in its linear regime. Bias is acquired through multiple instantaneous exposures with the telescope aperture sealed. Since this level can be different in each pixel, 15 different frames are combined using a median-stacking procedure to reduce possible fluctuations. This ensures that the uncertainty associated with the master bias frame is reduced to sufficiently small values to be neglected.

After bias subtraction, dark currents have to be considered: electrons can accumulate in the current from promotions to the conduction band due to thermal excitation within the detector material. Cooling the camera, as implemented in TOBi, can reduce this effect, but it remains relevant for longer exposures. As before, dark frames are acquired with the telescope aperture sealed, using integration times matching the science observations: 300 seconds for r-band and 1200 seconds for $H\alpha$ measurements. Again, a median of 15 different exposures is taken as the master value per pixel to suppress possible fluctuations, after subtracting the bias.

Last calibration step is the flat-field correction. It manifests as different response across detector pixels, producing a non-uniform field. Ten different exposures are acquired using a uniform illumination source, a lamp located on the telescope's cap used to illuminate the primary mirror. Again, all these frames, bias subtracted, are combined using a median, with an additional normalization for their median value.

Therefore the raw image is calibrated as:

$$Image = \frac{(Raw\ image - (Bias + Dark))}{Flat}. \quad (2)$$

2.2.2 Sky subtraction

All images contain the contribution from the sky background photons, which must be removed. The sky produces an approximately uniform signal across the field, shifting pixel values toward higher counts, while astronomical sources appear as the high-count tail of the pixel-value distribution. As a first-order approximation, the sky level can be estimated from the median of the image, after excluding the bright tail of the histogram. A more refined subtraction is then performed: sources such as stars and the galaxy itself are masked, and the background is computed on the residual image using the *Background* function from SEP library. SEP allows to construct a smooth background map by choosing the grid size on which the local sky median is computed. Larger grid sizes produce a smoother map, reducing halos produced by stars and other sources. In this work, the parameters used are a box of 64×64 pixels and a smoothing filtering

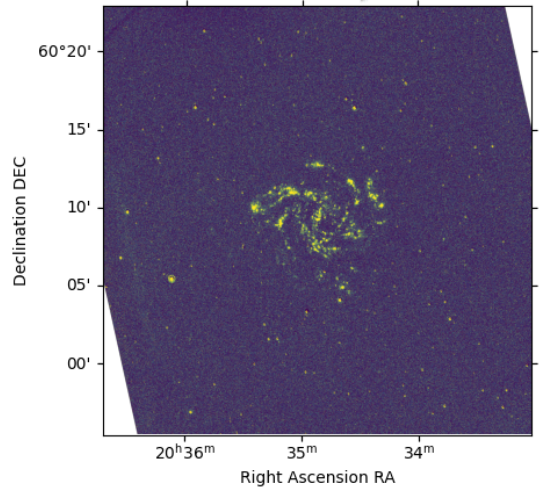


Figure 1. Final image of NGC 6946 from TOBi after all calibration steps.

window of size 15 for $H\alpha$ and 10 for r-band. In this process, a proper mask on sources is essential to avoid misidentifying them as sky, which would produce an oversubtraction and an underestimation of their flux.

Once the background subtracted image is obtained, the science frame is created by converting pixel values from ADU (analog to digital units) to electrons per second by multiplying by the gain ($G = 0.25 \frac{e^-}{ADU}$) and dividing by the exposure times. Then, to obtain the physical fluxes in $\frac{erg}{s \cdot cm^2}$, the values are multiplied by the telescope's calibration constant. The calibration constant is derived from objects with known physical flux: stars in the field of NGC 6946 are used, with theoretical fluxes taken from the GAIA catalog. For each star, the theoretical flux is divided by the measured one from TOBi (in electrons per second) within a circular aperture around the star, obtaining the conversion constant. For this purpose, four stars are selected: they lie in the central region of the field (not too close to the galaxy) where the flat field variations are minimal and are not saturated, to avoid flux loss. The best value of calibration constant is taken as the mean of the four individual values. This procedure is repeated on every image in both filters, obtaining a mean calibration constant for each band that is then applied to the data. Each exposure is taken at a different tilt angle to mitigate possible systematic errors due to potentially problematic pixels. Therefore, before combining the exposures with a median-stacking procedure, all images are reprojected onto a common world coordinate system.

The spectroscopic data, in contrast, are already reduced and noise subtracted, so no further calibration steps are required.

3 ANALYSIS

Given the data obtained as described in section 2, they are processed to extract the physical quantities of interest. In particular, the star formation rate, as discussed in section 1, is traced by H-alpha ($H\alpha$) flux. However, the $H\alpha$ filter is a narrow-band filter, meaning that the resulting signal includes not only the emission line flux but also part of the stellar continuum present in the r-band. To isolate the pure $H\alpha$ emission (Figure 1), the r-band continuum is subtracted from the H-alpha image with the relation:

$$f_{tot} = (f_{H\alpha} - C f_R) \Delta \lambda_{H\alpha}, \quad (3)$$

where f_{tot} represents the final flux per pixel, $f_{H\alpha}$ and f_R the fluxes per pixel respectively from the $H\alpha$ filter and the r-band in units of $\frac{erg}{s \cdot cm^2}$ and $\Delta\lambda_{H\alpha}$ is the bandwidth of the $H\alpha$ filter. C is a fudge factor calibrated by eye, chosen such that only the star forming regions are visible in the galaxy. In this case, $C = 1.4$.

After masking all sources outside the galaxy, the physical quantities are computed from this image. Elliptical apertures are taken from the center of the galaxy outward. For each of them, the flux is computed with *ApertureStat* module of *Astropy* with another median sky subtraction. Performing one additional sky evaluation guarantees a cleaner result, subtracting any possible residues from previous reductions. From the cumulative fluxes over increasing areas, the surface brightness (SB) is computed and the radial profiles of flux and SB are obtained.

The optimal flux is taken as the value for which the variation between two consecutive estimates is less than 0.5%. This gives an observed flux of $F_{obs} = (1.714 \pm 0.005) \times 10^{-11} \frac{erg}{s \cdot cm^2}$. The luminosity is then computed as:

$$L = 4\pi D^2 F, \quad (4)$$

where D the distance of the galaxy is $D = (7.72 \pm 0.32) Mpc$, giving $L = (1.22 \pm 0.11) \times 10^{41} \text{ erg/s}$. Next the SFR is obtained from:

$$SFR = K_{H\alpha} L, \quad (5)$$

using $K_{H\alpha} = 7.9 \times 10^{-42} \frac{M_{\odot}s}{erg \cdot yr}$; resulting in $SFR = (0.97 \pm 0.08) \frac{M_{\odot}}{yr}$.

The errors associated with the measurement have been estimated at pixel level. In particular, two main contributions are considered: the background noise, always present in the image, and the error on the photon counts from each source. The background error σ_b has been estimated as the standard deviation of the histogram distribution of all pixel values. Instead, the source error σ_{f_i} is derived assuming a Poisson distribution on the photons collected. So, from the flux of the pixel f_i in cgs, it is computed by reconverting it into electrons as:

$$\sigma_{f_i} [cgs] = \sqrt{\frac{\ll C \gg}{t_{exp} N_{exp}}} f_i [cgs], \quad (6)$$

where $\ll C \gg$ represents the mean calibration constant for $H\alpha$ and r images, t_{exp} the exposure time per filter and N_{exp} the number of exposures done in each filter. When both background and sources contribute to a pixel, the total error is $\sigma = \sqrt{\sigma_b^2 + \sigma_{f_i}^2}$. The error maps for the $H\alpha$ and r band are then combined using equation 3. Then, when computing the flux, the error associated with it is obtained as well. From this value, the uncertainties on luminosity and SFR are obtained with error propagation.

The value of the SFR has to be corrected for two processes, both evaluated from spectroscopy:

- the fact that H-alpha filter is not centered exclusively on $H\alpha$ emission line and therefore collects additional flux from nearby lines;
- dust attenuation.

In the first case, since the filter is not perfectly selective, the $H\alpha$ band also includes, after subtracting the sky level (done with a median in the regions near the lines), the contribution from the nitrogen-II emission lines. These two line fluxes, one bluer and one redder than H-alpha, have to be deducted for the estimate of the real flux, done with relation 7:

$$F_{H\alpha} = \frac{I_{H\alpha}}{I_{tot}} F_{obs}, \quad (7)$$

where $I_{H\alpha}$ is the integrated $H\alpha$ line, I_{tot} the total integrated flux in

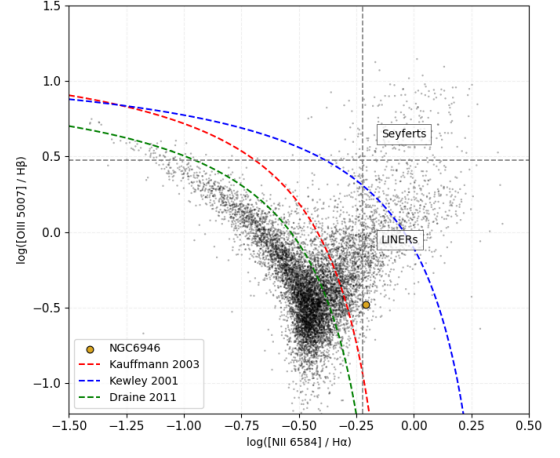


Figure 2. BPT diagnostic diagram using 10000 SDSS galaxies (gray points) and the spectroscopic measurements of NGC 6946 (yellow point). The theoretical curves from Kewley, Kauffman and Draine are shown too, with Seyferts and LINERs regions.

the band and F_{obs} the observed flux obtained before. The integrated line fluxes come from fitting a Gaussian profile over the spectroscopic line, multiplying by the transmission curve of the $H\alpha$ filter and integrating the result.

As discussed in section 1, dust absorbs part of the radiation emitted by young stars, reducing the $H\alpha$ detectable flux and re-emitting the absorbed energy at longer wavelengths. To correct for this effect, the Balmer Decrement technique is applied. This method uses the ratio of the H-alpha and the H-beta line as a tracer of dust: in absence of dust, the intrinsic ratio is $\frac{H\alpha}{H\beta} > 2.86$, while, if dust is present, it absorbs more in the region of the $H\beta$ and re-emits in the $H\alpha$ one leading to a different ratio. From this difference, the reddening can be reconstructed. As done for the step before, the line fluxes for $H\alpha$ and $H\beta$ are obtained from the spectrum by integrating the local background subtracted Gaussian fits of the two emissions. The reddening is then computed as:

$$E(B - V) = 1.97 \log_{10} \left[\frac{\left(\frac{H\alpha}{H\beta} \right)_{obs}}{2.86} \right], \quad (8)$$

with $\left(\frac{H\alpha}{H\beta} \right)_{obs}$ the ratio of the integrated lines. The attenuation $A_{H\alpha}$ is obtained with:

$$A_{H\alpha} = 3.33 \times E(B - V). \quad (9)$$

Finally, the total intrinsic flux $F_{H\alpha}^0$ is found by inverting the Balmer Decrement relation:

$$A_{H\alpha} = 2.5 \log_{10} \left(\frac{F_{H\alpha}^0}{F_{H\alpha}} \right), \quad (10)$$

where $F_{H\alpha}$ is the flux coming from the image and corrected with Equation 7. The flux corrected for dust absorption is now $F = (2.68 \pm 0.01) \times 10^{-10} \frac{erg}{s \cdot cm^2}$. Given this value, the luminosity and SFR are recomputed via Equations 4 and 5, yielding: $L_{H\alpha} = (1.81 \pm 0.01) \times 10^{42}$ and $SFR = (14.27 \pm 0.04) \frac{M_{\odot}}{yr}$. To determine the position of NGC 6946 in the BPT diagram, the intensities of the nitrogen (NII at $\lambda_{NII} = 658.3nm$) and oxygen ($OIII$ at $\lambda_{OIII} = 500.7nm$) lines are needed. In particular, the NII lies again in the $H\alpha$ filter and the integrated value is obtained using the same procedure as before: background subtraction, Gaussian fitting and integration.

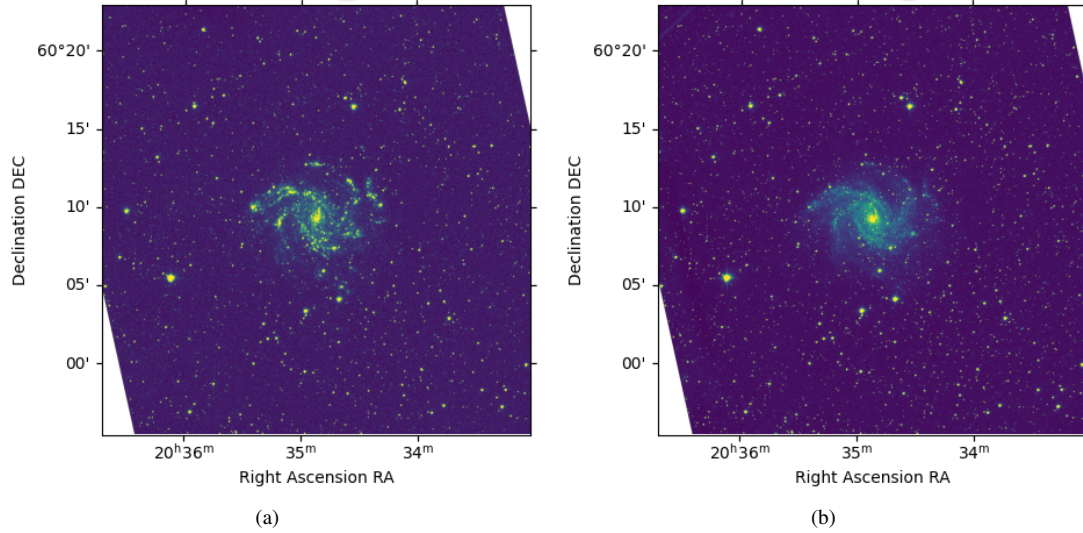


Figure 3. Left: master $H\alpha$ image of NGC 6946 obtained from median-stacking procedure. Right: master r-band image used to subtract the stellar continuum.

The same process is done for the $OIII$ line, but the sky baseline is subtracted for its local values, computed as the median of points before and after the line. Using these measurements with the $H\alpha$ and $H\beta$ lines computed before, the estimates for the diagram are obtained: $\log\left(\frac{NII}{H\alpha}\right) = -0.21$ and $\log\left(\frac{OIII}{H\beta}\right) = -0.48$. To populate the BPT diagram 10000 galaxies from the SDSS catalog are plotted together with NGC 6946 (Figure 2).

In the diagram, the theoretical behavior and regions are also plotted. In particular, the theoretical laws of Kewley, Kauffman and Draine divide the diagram in two main regions: below the curves, ionization is dominated by young stars photons; while above them, AGNs are the primary source of ionization. A further subdivision is considered: the region of Seyferts and LINERs (Low-Ionization Nuclear Emission-line Regions) galaxies. Seyferts exhibit strong ionization in both oxygen and nitrogen, originating from AGNs. LINERs instead show weak oxygen ionization but strong nitrogen one; in this case coming mostly from photoionization by evolved stars, with only a small fraction from a possible weak AGN radio feedback. NGC 6946 lies in the LINERs region, positioned between the theoretical models.

4 DISCUSSION

The SFR obtained for NGC 6946 after applying the dust correction confirms that the galaxy is star forming as expected. However, the derived value is higher than the theoretical estimate of $SFR \approx 3 \frac{M_{\odot}}{yr}$, reported in PHANGS- $H\alpha$ catalog (Leroy et al. (2019)). This overestimate likely arises from an incorrect dust attenuation correction: the spectroscopic data used for the Balmer decrement were taken in the center of the galaxy, where dust content is higher. This leads to an artificially large attenuation and therefore an overestimation of the $H\alpha$ flux. In contrast, the star forming regions are in the arms of the galaxy, where the contribution from dust is lower. Thus, a more reliable estimate of the star formation rate can be obtained by taking spectroscopic measurements in these outer regions. To test this hypothesis, an estimate can be made using the dust attenuation value reported in Kessler et al. (2020) of $A_{H\alpha} = 1.5\text{mag}$. This value corresponds to a median attenuation measured in 1934 apertures dis-

tributed around the galaxy. Using this attenuation and TOBi flux after the correction for the nitrogen contribution, the resulting SFR found is $SFR \approx 2.36$. This value is more consistent with the expected one, underlining the problem with the spectroscopic dataset.

Regarding the position of NGC 6946 on the BPT diagram, the galaxy appears to lie in an intermediate region depending on the classification curve adopted. It appears to fall within the LINERs, which can be plausible considering the high number of supernova events occurring in the galaxy. These supernova remnants can contribute to low-ionization emission and shift the line ratios toward the LINERs region. The fact that NGC 6946 does not appear clearly as a star forming galaxy can again likely be due to the location of the spectroscopic slit: the center hosts less star formation and more dust, which alters the values of the ratios leading to a misclassification.

5 CONCLUSIONS

NGC 6946, from TOBi observations, appears to be a star forming galaxy. However, the derived value of SFR is likely overestimated due to an incomplete analysis of dust attenuation. The BPT diagram is also not entirely conclusive, but it indicates that the main ionization process of the galaxy is stellar photoionization. A better analysis of both the SFR and the position in the BPT diagram could be done in the future with additional spectroscopic observations, focusing in particular on the outer regions of the galaxy.

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